How Nature's Language Is Being Translated In Biomimetic Technology

A product designed using biomimetic technology is not only innovative but also potentially sustainable. Nature works as a system that gives effective and sustainable outcomes

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"Learn from nature" — William Wordsworth

The world's principal mentor—Nature, has inspired many great creations starting from Velcro and climbing pads to genetic algorithms and manipulating robots that have led to a new era of technology. In biomimetics, the methods, mechanisms, and processes used in nature are adapted in multidisciplinary fields of engineering systems and technology to solve problems. In short, many of our technologies now try to mimic nature. Biomimetics is also referred to as bionics, biognosis, bioinspiration, and biomimicry.

The evolution of time has taught us what is capable, what is applicable, and what survives. Scientists and engineers prefer natural biological designs because these are highly practical, durable, efficient, and long-lasting with less energy consumption. They cut material costs, redefine and eliminate waste, thus driving more revenue to manufacturers.

Leonardo da Vinci, a universal genius, called nature as "mistress of all masters." He was inspired by birds and dreamt about flying but was unable to bring his idea into practice. However, his dream came true after five centuries through the Wright brothers' prototype that mimicked the flying birds.

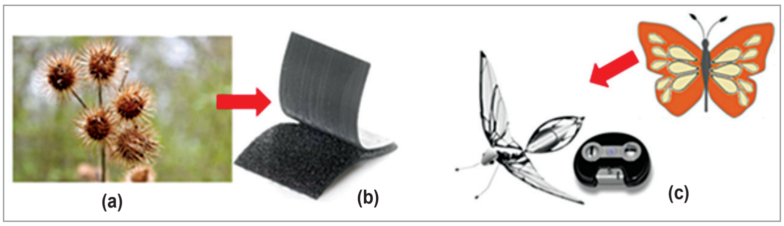

Fig. 1: (a) Burdock seeds (Courtesy: www.almanac.com), (b) Velcro (Courtesy: amazon.com), (c) Metafly (Courtesy: amazon.com)

Taking cue from the natural processes, humans have long turned to flora and fauna for inspiration, and have been translating the natural principles to engineering designs to tackle real-life challenges. These biomimetic products are bringing changes to the world and making our life a lot easier. Biomimetic technology is catalysing exponential economic growth and driving the global market. This article focuses on some devices that use biomimetic technology.

Smellicopter—the odour detecting unmanned aerial vehicle

We all have some idea about the biomimetic ornithone 'Metafly' (Fig. 1c)—a distinctive fascinating airborne device confronting many challenges. Now, similar bio-inspired research in robotics has led to 'smellicopter' (Fig. 2)—a hand-held drone that uses a live moth antenna to navigate toward odours. It has been designed collaboratively by researchers from the University of Washington (UW) and the Air Force Research Laboratory, USA. The moth's antenna, called electroantennogram (EAG), serves as the chemical sensor. The size of this drone is only nine centimetres.

The moth chosen for this research, Manduca sexta, also known as tobacco hornworm, is a voracious leaf-feeding pest, especially at its larval stage, and feeds on tobacco leaves and defoliates the plant. It